

Solutions to Problem Solving Class 2: Flow and Error Control + Multiplexing

Computer Communications – DE
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1. In Go-Back-N ARQ, if time-out expires for frame $n - k$, and the sender is currently sending frame n , frames $n - k, n - k + 1, \dots, n - 1, n$ have to be retransmitted. In Selective Repeat ARQ on the other hand, the receiver indicates what frame that should be retransmitted.
2. To include acknowledgement for one frame in the data frame of another frame.
3. **Stop-and-wait:** Expected number of next frame.
Go-back-n: Ack n means ack of frames $0, \dots, n - 1$, and that frame n is awaited.
Selective repeat: Ack n means ack of frames $0, \dots, n - 1$, and that frame n is awaited.
4. $2^m - 1 = 15, m = 4$
5. $\frac{2^7}{2} = 64$
6. 01234567012345670123
7. See Figure 1 a).
8. See Figure 1 b).
9. Specified frames are retransmitted.
10. By time-out.
11. If, for example, we choose $m=2$, and a window size of 4, the following can happen. Assume that all acknowledgements are lost, and the frame 0 timeout expires. Then, the sender retransmits frame 0. But the receiver, which has received frames 0,1,2, and 3, is now expecting a new frame 0. So when the retransmitted frame 0 arrives, it is incorrectly assumed to be a new frame.
12. The bandwidth-delay product is a measure of the number of bits we can send out of our system while waiting for news from the receiver. A high bandwidth-delay product imposes a large window.
13. 2500Hz

14. a) 1.44 Mbps

b) 30

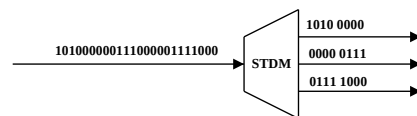
15. a) 101111101 1 000111110 0 001100111 1 111000101 0 →

b) 1 Mbps, 1 frame == 10 bits

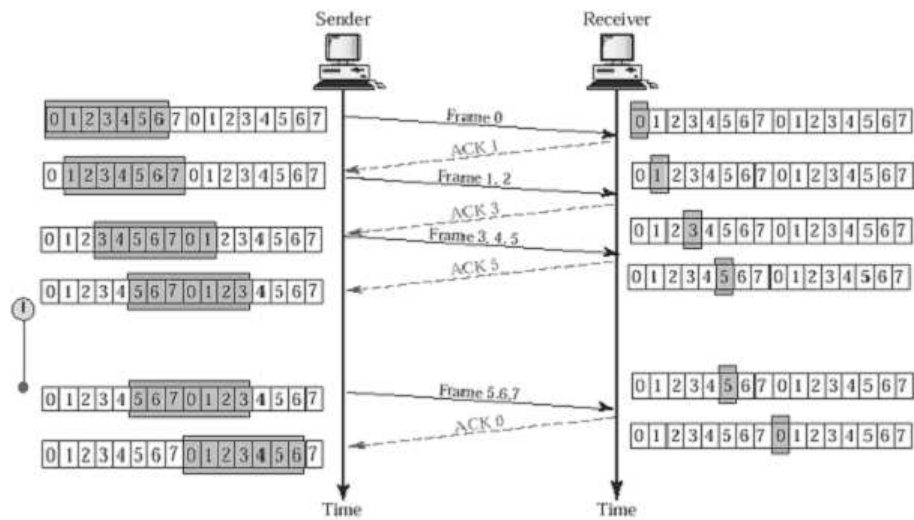
c) 1 microsecond

d) 10 microseconds, 1 frame == 10 bits

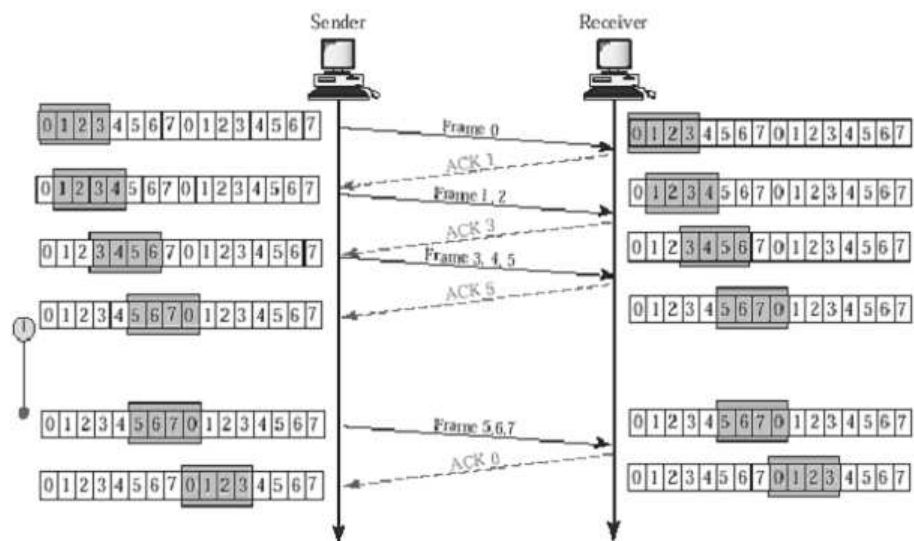
16. a)



b) 3 Mbps



a)



b)

Figure 1: a) Exercise 7, b) Exercise 8.